

Sathwik Mohan Vallala

Herndon, VA, USA | 571-373-6079 | sathwikv143@gmail.com | in/sathwik-mohan | H1B(Oct.2028)/I-140 Approved

SUMMARY

Cybersecurity professional with over 6 years of experience breaking and securing business-critical products through offensive security, vulnerability management, and shift-left application security. Experienced in STRIDE threat modeling, vulnerability assessment, penetration testing, OWASP-aligned testing, and secure code review to identify attack paths early and guide remediation. Skilled at incorporating SAST, DAST, and SCA into CI/CD workflows and applying AI and AI Security practices to automate analysis, accelerate triage, and reduce risk. Collaborates closely with engineering teams to strengthen secure design and deliver resilient products; holds the Certified Ethical Hacker (CEH) certification.

EXPERIENCE

Product Security Engineer

Jun. 2024 – Jul. 2026

Omnissa, LLC

Herndon, VA

- Integrated AI and AI Security into vulnerability management workflows, using CVSS to prioritize and accelerate remediation of 50+ critical vulnerabilities.
- Performed OWASP-aligned vulnerability assessments and attack surface analysis for 30+ customer features, combining AI-assisted analysis with manual validation to deliver secure implementation recommendations.
- Applied AI-assisted offensive security and penetration testing across 10+ product releases, accelerating reconnaissance, payload generation, and vulnerability triage.
- Shifted security left across 20+ releases by incorporating secure code review and SAST, DAST, and SCA results into CI/CD workflows while preserving high-confidence findings.
- Applied STRIDE threat modeling to identify attack paths and validate mitigations for 10+ major engineering initiatives, guiding secure design decisions early in the SDLC.
- Expanded AI-assisted static analysis and code understanding across 75% of product codebases, identifying risks earlier and scaling secure code review coverage.
- Presented complex security issues to 5+ cross-functional teams, strengthening security awareness and alignment on product risk.
- Founded and facilitated a monthly cybersecurity roundtable focused on emerging threats and industry events, strengthening team awareness and knowledge sharing.

Product Security Engineer (CarbonBlack)

Nov. 2023 – Jun. 2024

Broadcom Inc.

Herndon, VA

- Performed vulnerability assessments that identified 10 product security vulnerabilities, using CVSS-based vulnerability management to prioritize remediation and reduce attack surface.
- Conducted OWASP-aligned offensive security and penetration testing across 4 product releases, applying DAST techniques to identify, validate, and exploit vulnerabilities.
- Applied STRIDE threat modeling to shift left by identifying attack vectors early and guiding secure design decisions with engineering teams.
- Performed SAST and SCA informed secure code reviews within CI/CD workflows to identify and mitigate exploitable risks.
- Translated complex technical findings into clear risk context, building understanding and remediation buy-in across product teams.

Security Response Engineer

Apr. 2022 – Nov. 2023

VMWare Inc.

Herndon, VA

- Managed 50+ incoming vulnerability reports, triaging findings, assessing urgency, and driving timely response through the vulnerability management lifecycle.
- Conducted vulnerability assessments for reported issues, assigned CVSS v3.1 severity scores, and documented exploitation steps and mitigation strategies.
- Coordinated vulnerability response across internal teams and planned clear, timely customer communications.
- Partnered with security researchers on responsible disclosure and published 3 VMware Security Advisories (VMSAs) detailing required customer actions.
- Defined and implemented mitigations, indicators of compromise (IOCs), workarounds, and remediations to reduce customer exposure.
- Monitored for active exploitation and updated security advisories as threat conditions evolved.
- Developed and implemented security-response automations that streamlined team workflows and improved operational efficiency.

Digital Forensic Researcher & Developer

Dec. 2020 – Mar. 2022

ADF Solutions Inc

Reston, VA

- Improved the accuracy and reliability of digital artifact collection modules through rigorous testing and verification.
- Independently researched and developed new digital artifact collection modules, expanding the system's evidence-collection capabilities.
- Designed collection approaches for 20+ digital artifacts, resolving complex evidence-acquisition challenges.
- Collaborated across the development team and adapted quickly to varied project requirements and investigative use cases.
- Built 10+ Python plugins to parse and extract digital artifacts and evidence, streamlining investigative workflows.
- Performed quality assurance, testing, and validation on developed plugins before integration into the operational DEI Tool.

Trainee Security Analyst - Internship

SOC Experts

Jul. 2019 – Oct. 2019

Bangalore, India

- Used three SIEM platforms—Splunk, IBM QRadar, and LogRhythm—to support threat detection and incident response in a simulated SOC.
- Monitored and analyzed security logs to identify suspicious activity and potential incidents.
- Conducted foundational digital forensics by collecting and analyzing evidence for security investigations.
- Collaborated within a simulated SOC to identify and respond to security threats.

Security Analyst - Internship

Berry9 IT Services Pvt Ltd

Jun. 2017 – Jun. 2018

Hyderabad, India

- Developed hands-on ethical hacking skills through self-directed security projects and practical testing.
- Identified vulnerabilities during a real-world healthcare penetration testing engagement.
- Earned the Certified Ethical Hacker (CEH) credential, validating foundational offensive security knowledge.

EDUCATION

George Mason University

Master's of Science in Digital Forensics & Cyber Analysis - 3.89/4.0

Fairfax, VA

Jawaharlal Nehru Technological University

Bachelor's of Technology in Computer Science Engineering - 4.0/4.0

Hyderabad, India

PROJECTS

Collaborative ToDo | NodeJS, VibeCoding

Apr. 2026 – May 2026

- Built a collaborative task management application using AI-assisted coding workflows
- Deployed the application on self-hosted infrastructure to enhance privacy and data control
- Implemented secure passwordless authentication for streamlined user access
- A shared to-do lists that enable multiple users to create, manage, and complete tasks collaboratively

Autopsy WhatsApp Plugin | Python, Forensics, Evidence

Apr. 2020 – May 2020

- Enhanced *Autopsy* capabilities by independently developing a new plugin, demonstrating expertise in forensics tool customization & automation.
- Recovered & parsed digital evidence from the *WhatsApp* application on *Windows* systems, utilizing advanced data extraction & parsing techniques.

Chat over network with Python | Python, Network, Encryption

Jun. 2018 – Jul. 2018

- Built a privacy-forward chat application with strong cryptography, empowering secure communication.
- Implemented end-to-end encryption for secure communication, ensuring message confidentiality & protecting user data.
- Enabled user control & independence by designing a self-hostable chat server, fostering privacy & data sovereignty.

File Sharing | PHP, HTML, SQL, Network

Mar. 2018 – Apr. 2018

- Streamlined internal file sharing with a user-friendly platform designed for specific network needs.
- Prioritized security by implementing password authentication to restrict unauthorized access to shared files.

CERTIFICATIONS & AWARDS

Security Code Review Ninja - 2nd Degree Black Belt | Secure Coding Dojo

December 2023

Digital Evidence Investigator | ADF Solutions

December 2020

Certified Autopsy User | Basis Technology

October 2020

Certified Ethical Hacker | EC Council

October 2017

TECHNICAL SKILLS

Security: LLMs, MCPs, AI Security, Vulnerability Assessment, Application Security, Threat Hunting, Penetration Testing, Incident Response, Secure Code Review, SAST, DAST, AWS Security, Log Analysis, Security Event Management, Ethical Hacking, Forensic Analysis, Malware Reverse Engineering, Network Forensics, Memory Forensics, Open Source Intelligence, Cloud Security, SaaS

Programming: Python, Bash, SQL, PHP, JavaScript

Tools: Coverity, Semgrep, Docker, Burp Suite, ZAP, OWASP, WinHex, OllyDbg, IDA, Ghidra, Snort, Zeek, IBM QRadar, LogRhythm, Metasploit, Linux, Nmap, Wireshark, Ettercap, Aircrack-ng, SQLMap, EnCase, FTK, Autopsy, BlackBag, X-Ways, AXIOM, Volatility, tcpdump, Falcon, hashing tools, write blockers, Splunk, Hydra, John the Ripper, Nikto, Nessus, open-source security tools